

ARYAN SATYENDRA KUMAR

Machine Learning | Motorsport Analytics | Full-Stack Data Products

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EDUCATION

B.Tech, Electronics and Communication Engineering, Minor in Computer Science

National Institute of Technology Goa - CGPA: 8.23/10

Expected 2027

Goa, India

MOTORSPORT ANALYTICS

F1 Race Intelligence - Probabilistic Race Forecasting Platform

Python, FastF1, scikit-learn, Pandas, FastAPI, Next.js, TypeScript

Aug 2025 - Present

GitHub | Predictions

- Engineered a post-qualifying forecasting pipeline from FastF1 data. Trained a 1,200-tree Random Forest on finish-position delta with median imputation, one-hot encoding, out-of-bag evaluation, and per-race retraining.
- Designed leakage-safe rolling driver/constructor form and circuit-archetype features, blending historical priors with FP1/FP2/FP3/qualifying signals, weather, tyre degradation, pit loss, and SC/VSC likelihood.
- Backtested 70 Grands Prix from 2023-2025, achieving 58.6% winner accuracy and 79.7% top-10 overlap while reducing finish-position MAE by 5.2% versus a grid-order baseline (3.14 vs. 3.31 positions).
- Published 2026 pre-race winner forecasts with only two incorrect calls to date. Built baseline comparisons, Brier scoring, calibration analysis, leakage audits, grid-model blend search, and versioned prediction logging.
- Generated 68% and 95% intervals plus Monte Carlo win, podium, top-10, and rank probabilities. Served forecasts, feature importance, driver comparisons, and a grid simulator through FastAPI and Next.js.

EXPERIENCE

Project Intern - Mobile Air Quality Intelligence

Indian Institute of Technology Madras - Prof. Raghunathan Rengaswamy

May 2026 - Jul 2026

Chennai, India

- Developed a multimodal roadside PM_{2.5} pipeline combining mobile sensors, OPC particle bins, MERRA-2/CAMS atmospheric variables, meteorology, GPS/OSM context, and camera imagery.
- Fine-tuned YOLO vehicle detectors and applied SegFormer road segmentation to derive traffic counts, road exposure, texture, brownness, and dust-related visual proxies.
- Built ResNet50-GRU and tree-based models for background pollution and local roadside increments using leakage-aware whole-date validation, with filter-sampling and XRF validation against physical particulate measurements.

Signalling and Telecommunications Intern

South Western Railways, RVNL

May 2025 - Jul 2025

Goa, India

- Analysed Kyosan Electronic Interlocking in a safety-critical railway environment, tracing system feedback paths, interlocking logic, relay interfaces, and failure-mode handling.

SELECTED PROJECTS

NyayaSetu - Multilingual ML-RAG Assistant for Legal Aid

FastAPI, Next.js, XLM-R, Sentence Transformers, ChromaDB, Groq API

Jun 2026

GitHub

- Fine-tuned XLM-R classifiers for legal issue and urgency detection. Built an official-source RAG workflow with citation-grounded answers, guardrails, explainability, and retrieval evaluation using Recall@1/3/5 and MRR.

HostelMitra - Hostel Complaint Management Portal

Firebase, Firestore, Google OAuth

2025

GitHub | Live Demo

- Built and deployed a portal used by NIT Goa students with role-based access, real-time complaint tracking, voting, admin resolution, and timestamped audit trails.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, Java, TypeScript **ML/Data:** PyTorch, scikit-learn, Pandas, NumPy, Random Forests, feature engineering, temporal backtesting, model calibration, uncertainty modelling

Vision/NLP: YOLO, SegFormer, CNNs, GRUs, transformers, XLM-R, embeddings, RAG, ChromaDB **Engineering:** FastAPI, REST APIs, Next.js, Firebase, PostgreSQL, Git, Docker

ACHIEVEMENTS & LEADERSHIP

- Selected for Amazon ML Summer School 2026 and ranked 251 of 7,000+ teams in the Amazon ML Challenge 2025.
- Finalist at India Innovates 2026 and in national case-study competitions hosted across IIM, IIT, and NIT platforms.
- GDG Organizer and AI/ML Lead, leading a 100-participant hackathon and delivering ML and DSA workshops.
- Captain, Institute Badminton Team, and former national-level badminton player.

CERTIFICATIONS

Machine Learning Specialization - Andrew Ng | Deep Learning Specialization - Andrew Ng | Google Data Analytics Professional Certificate